

RESEARCH STUDY - FEBRUARY 2026

AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors:

US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategic and Economic Analysis - ANNEX B - WORKING PAPER

**76.9% of global operational AI
compute = USA**

**1.59x energy cost EU/US (PPP-
adjusted)**

**3.46:1 CACI US/EU ratio (Power
Mode)**

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ANNEX B - WORKING PAPER

The Compute-Adjusted Competitiveness Index (CACI): Measuring the Impact of American AI Protectionism on Global AI Competitiveness

This annex presents the working paper formalizing the CACI index in the format of an economic journal article. It introduces the Compute-Adjusted Competitiveness Index (CACI), a novel composite indicator designed to measure national AI competitiveness by capturing the interaction between installed computing capacity, energy costs, GDP, and AI workforce. JEL Codes: F13 (Trade Policy), L63 (Semiconductors), O33 (Technological Change), O38 (Public Policy). Keywords: AI competitiveness, compute gap, technological protectionism, export controls, CACI, panel data, semiconductors, energy policy, Section 232, European digital sovereignty, Cloud Sovereignty Mandates, Phys/Sov decomposition.

B.1 Introduction

Artificial intelligence is reshaping the foundations of global economic competitiveness. Since the launch of ChatGPT in November 2022 and the subsequent wave of investment in foundation models, generative AI has emerged as a transformative cross-cutting technology. Yet access to the infrastructure necessary to train and

deploy frontier models—particularly advanced GPUs and affordable energy for data centers—has become profoundly asymmetric.

In this context, the United States has progressively established a regime of control over access to cutting-edge AI technologies. Starting in October 2022, the Bureau of Industry and Security (BIS) imposed restrictions on exports of advanced GPUs to China. In January 2025, the AI Diffusion Rule segmented the world into three access tiers. In May 2025, the Trump administration repealed this rule and replaced it in January 2026 with a final rule combining 25 percent tariffs (Section 232) on advanced AI semiconductors with revised export controls.

These measures, officially motivated by national security imperatives, de facto produce a structural competitive advantage for American companies: they enjoy unlimited access to frontier compute, while actors in other regions—including European allies—face increasing constraints in terms of cost, availability, and regulatory certainty.

Despite the magnitude of these developments, economic literature lacks a quantitative framework to measure the resulting competitiveness gap. Existing indicators—R&D spending, number of patents, AI publication metrics—do not capture the hardware infrastructure that increasingly determines productive capacity in AI. This article fills this gap by proposing the Compute-Adjusted Competitiveness Index (CACI), a composite indicator that integrates installed computing capacity, energy cost, GDP, and AI workforce into a single analytical framework.

B.2 Literature Review

The foundational intuition of our framework derives from the theory of general-purpose technologies (GPTs) by Bresnahan and Trajtenberg (1995). GPTs are characterized by their pervasiveness, inherent potential for improvement, and innovation complementarities. AI—particularly large language models and foundation models—clearly meets these criteria. However, unlike previous GPTs (steam, electricity, semiconductors), AI requires massive computing infrastructure whose cost and distribution are highly unequal.

Brynjolfsson, Rock, and Syverson (2019) provide the crucial complement with their J-curve theory: productivity gains from GPTs are delayed because firms must invest in complementary assets—organizational restructuring, worker training, process redesign—before reaping the benefits. This implies that countries with early access to compute enjoy doubly increasing returns: they start the J-curve earlier and accumulate complementary assets that laggards cannot easily replicate.

B.2.2 Instrumentalized Interdependence and Control of Chokepoints

Farrell and Newman (2019) introduce the concept of instrumentalized interdependence: states can exploit the asymmetric structure of global networks to constrain other actors. They identify two mechanisms: the panoptic effect (surveillance via control of information nodes) and the chokepoint effect (disruption via control of supply bottlenecks). Nvidia's design monopoly (over 80% of AI training GPUs) and ASML's monopoly in EUV lithography constitute two exploitable chokepoints.

The October 2022 export controls, the AI Diffusion Rule, and Section 232 explicitly instrumentalize these chokepoints. The Trump administration proposed monetizing this leverage (25% of Chinese sales revenues, September 2025), marking a transition from access denial to rent extraction—a theoretically significant evolution that national security frameworks alone cannot explain.

B.3 The CACI Framework

We argue that national AI competitiveness is determined by four interacting factors: (i) installed computing capacity F , (ii) energy cost for data centers E , (iii) AI-skilled labor L , and (iv) regulatory access R (BIS Tier 1/2/3 classification). The CACI captures their interaction in a geometric multiplicative structure that reflects complementarities between factors.

The weights are $F = 0.40$ (compute, dominant), $L = 0.20$ (human capital), $R = 0.15$ (regulatory access), $E = 0.25$ (energy cost, denominator).

B.3.4 Phys/Sov Extension

Experience documented in Chapter VI ter (UAE case: 99.6% of Emirati F_{total} held by US-side operators) reveals that simple physical location of compute can mask a critical jurisdictional dependency. To address this dimension, we decompose F into two multiplicative components: $F(r,t) = F_{\text{phys}}(r,t) \times F_{\text{sov}}(r,t)$, where F_{phys} is physically installed compute and F_{sov} is the fraction of F_{phys} held by non-US operators.

B.6 The US-EU Compute Gap

The CACI framework allows for precise quantification of compute advantages between countries. On the April 2026 snapshot, the US/EU(13) raw operational installed compute ratio stands at 17.6:1, translated by the geometric Power Mode formula into a US/EU CACI ratio of 3.46:1.

The structural gap is driven by a three-tier protectionist architecture: (1) Export controls (access denial), (2) Section 232 tariffs (cost surtax), and (3) Capitalistic gravity (energy cost 1.59x lower in the US, capex concentration). A fourth tier, the Cloud Sovereignty Mandates (2028), threatens the sovereign layer of cloud workloads.

B.9 Conclusion

This article introduced the CACI as the first quantitative framework for measuring AI competitiveness through compute infrastructure. Econometric validation ($N=60$, 2020-2024) demonstrates that CACI is a significant predictor of AI productivity (elasticity 0.251, $p<0.01$). The structural gap of 3.46:1 cannot be closed by conventional industrial policy alone; it requires targeted strategic autonomy on energy, sovereign cloud, and strategic GPU reserves.

Table B.1. Panel variables and sources.

Source: Author's compilation. CACI is calculated according to the geometric Power Mode formula.

Variable	Definition	Unit	Source
$F(r,t)$	Installed accessible AI	PetaFLOPs (H100-eq)	Epoch AI, Hawkins et al.

	FLOPs		(2025)
$E(r,t)$	Data center energy cost (PPP-adj)	USD/MWh	Eurostat, EIA, IEA (2025)
$L(r,t)$	AI Workforce (STEM + certs proxy)	Thousands	OECD, LinkedIn Graph
$R(r,t)$	Regulatory access (Tier 1/2/3)	Index 0-1	BIS, Section 232
$PROD(r,t)$	AI-intensive sector productivity gain	Annual % gain	McKinsey, IMF, Fed Board

Notes

1. Bresnahan, T.F. & Trajtenberg, M. (1995). General purpose technologies: Engines of growth?, Journal of Econometrics.
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3. Farrell, H. & Newman, A. (2019). Weaponized interdependence.
4. IEA (2025), 'Energy and AI', Special Report. Energy cost differential 1.59x after PPP adjustment.

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